Everyday Math

481. One elementary school in Ohio has 684 students and 45 teachers. Write the student-to-teacher ratio as a unit rate.

482. The average American produces about 1,600 pounds of paper trash per year (365 days). How many pounds of paper trash does the average American produce each day? (Round to the nearest tenth of a pound.)

483. A popular fast food burger weighs 7.5 ounces and contains 540 calories, 29 grams of fat, 43 grams of carbohydrates, and 25 grams of protein. Find the unit rate of (a) calories per ounce (b) grams of fat per ounce (c) grams of carbohydrates per ounce (d) grams of protein per ounce. Round to two decimal places.

484. A 16-ounce chocolate mocha coffee with whipped cream contains 470 calories, 18 grams of fat, 63 grams of carbohydrates, and 15 grams of protein. Find the unit rate of (a) calories per ounce (b) grams of fat per ounce (c) grams of carbohydrates per ounce (d) grams of protein per ounce.

Writing Exercises

485. Would you prefer the ratio of your income to your friend's income to be 3/1 or 1/3? Explain your reasoning.

486. The parking lot at the airport charges $0.75 for every 15 minutes. (a) How much does it cost to park for 1 hour? (b) Explain how you got your answer to part (a). Was your reasoning based on the unit cost or did you use another method?

487. Kathryn ate a 4-ounce cup of frozen yogurt and then went for a swim. The frozen yogurt had 115 calories. Swimming burns 422 calories per hour. For how many minutes should Kathryn swim to burn off the calories in the frozen yogurt? Explain your reasoning.

488. Mollie had a 16-ounce cappuccino at her neighborhood coffee shop. The cappuccino had 110 calories. If Mollie walks for one hour, she burns 246 calories. For how many minutes must Mollie walk to burn off the calories in the cappuccino? Explain your reasoning.

Self Check

(a) After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.

I can...	Confidently	With some help	No-I don't get it!
write a ratio as a fraction.			
write a rate as a fraction.			
find unit rates.			
find unit price.			
translate phrases to expressions with fractions.			

(b) After reviewing this checklist, what will you do to become confident for all objectives?

5.7 Simplify and Use Square Roots

Learning Objectives

By the end of this section, you will be able to:

- Simplify expressions with square roots
- Estimate square roots
- Approximate square roots
- Simplify variable expressions with square roots
- Use square roots in applications

☑ **BE PREPARED** 5.19 Before you get started, take this readiness quiz.

Simplify: $(-9)^2$.

If you missed this problem, review Example 3.52.

BE PREPARED 5.20 Round 3.846 to the nearest hundredth.
If you missed this problem, review Example 5.9.

BE PREPARED 5.21 Evaluate $12d$ for $d = 80$.
If you missed this problem, review Example 2.14.

Simplify Expressions with Square Roots

To start this section, we need to review some important vocabulary and notation.

Remember that when a number n is multiplied by itself, we can write this as n^2, which we read aloud as "n squared." For example, 8^2 is read as "8 squared."

We call 64 the *square* of 8 because $8^2 = 64$. Similarly, 121 is the square of 11, because $11^2 = 121$.

Square of a Number

If $n^2 = m$, then m is the square of n.

Modeling Squares

Do you know why we use the word *square*? If we construct a square with three tiles on each side, the total number of tiles would be nine.



This is why we say that the square of three is nine.

$$3^2 = 9$$

The number 9 is called a perfect square because it is the square of a whole number.

MANIPULATIVE MATHEMATICS

Doing the Manipulative Mathematics activity Square Numbers will help you develop a better understanding of perfect square numbers

The chart shows the squares of the counting numbers 1 through 15. You can refer to it to help you identify the perfect squares.

Number	n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Square	n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225

Perfect Squares

A **perfect square** is the square of a whole number.

What happens when you square a negative number?

$$\begin{aligned}(-8)^2 &= (-8)(-8)\\ &= 64\end{aligned}$$

When we multiply two negative numbers, the product is always positive. So, the square of a negative number is always positive.

The chart shows the squares of the negative integers from -1 to -15.

Number	n	–1	–2	–3	–4	–5	–6	–7	–8	–9	–10	–11	–12	–13	–14	–15
Square	n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225

Did you notice that these squares are the same as the squares of the positive numbers?

Square Roots

Sometimes we will need to look at the relationship between numbers and their squares in reverse. Because $10^2 = 100$, we say 100 is the square of 10. We can also say that 10 is a square root of 100.

Square Root of a Number

A number whose square is m is called a square root of m.

If $n^2 = m$, then n is a **square root** of m.

Notice $(-10)^2 = 100$ also, so -10 is also a square root of 100. Therefore, both 10 and -10 are square roots of 100.

So, every positive number has two square roots: one positive and one negative.

What if we only want the positive square root of a positive number? The *radical sign*, $\sqrt{\ }$, stands for the positive square root. The positive square root is also called the **principal square root**.

Square Root Notation

$\sqrt{m}$ is read as "the square root of m."

If $m = n^2$, then $\sqrt{m} = n$ for $n \geq 0$.

radical sign → $\sqrt{m}$ ← radicand

We can also use the radical sign for the square root of zero. Because $0^2 = 0$, $\sqrt{0} = 0$. Notice that zero has only one square root.

The chart shows the square roots of the first 15 perfect square numbers.

$\sqrt{1}$	$\sqrt{4}$	$\sqrt{9}$	$\sqrt{16}$	$\sqrt{25}$	$\sqrt{36}$	$\sqrt{49}$	$\sqrt{64}$	$\sqrt{81}$	$\sqrt{100}$	$\sqrt{121}$	$\sqrt{144}$	$\sqrt{169}$	$\sqrt{196}$	$\sqrt{225}$
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

EXAMPLE 5.69

Simplify: (a) $\sqrt{25}$ (b) $\sqrt{121}$.

Solution

(a)

	$\sqrt{25}$
Since $5^2 = 25$	5

(b)

	$\sqrt{121}$
Since $11^2 = 121$	11

TRY IT 5.137 Simplify: (a) $\sqrt{36}$ (b) $\sqrt{169}$.

TRY IT 5.138 Simplify: (a) $\sqrt{16}$ (b) $\sqrt{196}$.

Every positive number has two square roots and the radical sign indicates the positive one. We write $\sqrt{100} = 10$. If we want to find the negative square root of a number, we place a negative in front of the radical sign. For example, $-\sqrt{100} = -10$.

EXAMPLE 5.70

Simplify. (a) $-\sqrt{9}$ (b) $-\sqrt{144}$.

Solution

(a)

	$-\sqrt{9}$
The negative is in front of the radical sign.	-3

(b)

	$-\sqrt{144}$
The negative is in front of the radical sign.	-12

TRY IT 5.139 Simplify: (a) $-\sqrt{4}$ (b) $-\sqrt{225}$.

TRY IT 5.140 Simplify: (a) $-\sqrt{81}$ (b) $-\sqrt{64}$.

Square Root of a Negative Number

Can we simplify $\sqrt{-25}$? Is there a number whose square is -25?

$$(\quad)^2 = -25?$$

None of the numbers that we have dealt with so far have a square that is -25. Why? Any positive number squared is positive, and any negative number squared is also positive. In the next chapter we will see that all the numbers we work with are called the real numbers. So we say there is no real number equal to $\sqrt{-25}$. If we are asked to find the square root of any negative number, we say that the solution is not a real number.

EXAMPLE 5.71

Simplify: (a) $\sqrt{-169}$ (b) $-\sqrt{121}$.

Solution

(a) There is no real number whose square is -169. Therefore, $\sqrt{-169}$ is not a real number.

(b) The negative is in front of the radical sign, so we find the opposite of the square root of 121.

	$-\sqrt{121}$
The negative is in front of the radical.	-11

TRY IT 5.141 Simplify: (a) $\sqrt{-196}$ (b) $-\sqrt{81}$.

TRY IT 5.142 Simplify: (a) $\sqrt{-49}$ (b) $-\sqrt{121}$.

Square Roots and the Order of Operations

When using the order of operations to simplify an expression that has square roots, we treat the radical sign as a grouping symbol. We simplify any expressions under the radical sign before performing other operations.

EXAMPLE 5.72

Simplify: (a) $\sqrt{25}+\sqrt{144}$ (b) $\sqrt{25+144}$.

Solution

(a) Use the order of operations.

	$\sqrt{25}+\sqrt{144}$
Simplify each radical.	$5+12$
Add.	17

(b) Use the order of operations.

	$\sqrt{25+144}$
Add under the radical sign.	$\sqrt{169}$
Simplify.	13

TRY IT 5.143 Simplify: (a) $\sqrt{9}+\sqrt{16}$ (b) $\sqrt{9+16}$.

TRY IT 5.144 Simplify: (a) $\sqrt{64+225}$ (b) $\sqrt{64}+\sqrt{225}$.

Notice the different answers in parts (a) and (b) of Example 5.72. It is important to follow the order of operations correctly.

In ⓐ, we took each square root first and then added them. In ⓑ, we added under the radical sign first and then found the square root.

Estimate Square Roots

So far we have only worked with square roots of perfect squares. The square roots of other numbers are not whole numbers.

Number	Square root
4	$\sqrt{4}=2$
5	$\sqrt{5}$
6	$\sqrt{6}$
7	$\sqrt{7}$
8	$\sqrt{8}$
9	$\sqrt{9}=3$

We might conclude that the square roots of numbers between 4 and 9 will be between 2 and 3, and they will not be whole numbers. Based on the pattern in the table above, we could say that $\sqrt{5}$ is between 2 and 3. Using inequality symbols, we write

$$2 < \sqrt{5} < 3$$

EXAMPLE 5.73

Estimate $\sqrt{60}$ between two consecutive whole numbers.

Solution

Think of the perfect squares closest to 60. Make a small table of these perfect squares and their squares roots.

Number	Square root
36	6
49	7
64	8
81	9

60 → $\sqrt{60}$

Locate 60 between two consecutive perfect squares.	$49 < 60 < 64$
$\sqrt{60}$ is between their square roots.	$7 < \sqrt{60} < 8$

TRY IT 5.145 Estimate $\sqrt{38}$ between two consecutive whole numbers.

TRY IT 5.146 Estimate $\sqrt{84}$ between two consecutive whole numbers.

Approximate Square Roots with a Calculator

There are mathematical methods to approximate square roots, but it is much more convenient to use a calculator to find square roots. Find the $\sqrt{\ }$ or $\sqrt{x}$ key on your calculator. You will need to use this key to approximate square roots. When you use your calculator to find the square root of a number that is not a perfect square, the answer that you see is not the exact number. It is an approximation, to the number of digits shown on your calculator's display. The symbol for an approximation is $\approx$ and it is read *approximately*.

Suppose your calculator has a 10-digit display. Using it to find the square root of 5 will give 2.236067977. This is the approximate square root of 5. When we report the answer, we should use the "approximately equal to" sign instead of an equal sign.

$$\sqrt{5} \approx 2.236067978$$

You will seldom use this many digits for applications in algebra. So, if you wanted to round $\sqrt{5}$ to two decimal places, you would write

$$\sqrt{5} \approx 2.24$$

How do we know these values are approximations and not the exact values? Look at what happens when we square them.

$$\begin{aligned} 2.236067978^2 &= 5.000000002 \\ 2.24^2 &= 5.0176 \end{aligned}$$

The squares are close, but not exactly equal, to 5.

EXAMPLE 5.74

Round $\sqrt{17}$ to two decimal places using a calculator.

Solution

	$\sqrt{17}$
Use the calculator square root key.	4.123105626
Round to two decimal places.	4.12
	$\sqrt{17} \approx 4.12$

TRY IT 5.147 Round $\sqrt{11}$ to two decimal places.

TRY IT 5.148 Round $\sqrt{13}$ to two decimal places.

Simplify Variable Expressions with Square Roots

Expressions with square root that we have looked at so far have not had any variables. What happens when we have to find a square root of a variable expression?

Consider $\sqrt{9x^2}$, where $x \geq 0$. Can you think of an expression whose square is $9x^2$?

$$\begin{aligned} (?)^2 &= 9x^2 \\ (3x)^2 &= 9x^2 \qquad \text{so } \sqrt{9x^2} = 3x \end{aligned}$$

When we use a variable in a square root expression, for our work, we will assume that the variable represents a non-negative number. In every example and exercise that follows, each variable in a square root expression is greater than or equal to zero.

EXAMPLE 5.75

Simplify: $\sqrt{x^2}$.

Solution

Think about what we would have to square to get x^2. Algebraically, $(?)^2 = x^2$

	$\sqrt{x^2}$
Since $(x)^2 = x^2$	x

TRY IT 5.149 Simplify: $\sqrt{y^2}$.

TRY IT 5.150 Simplify: $\sqrt{m^2}$.

EXAMPLE 5.76

Simplify: $\sqrt{16x^2}$.

Solution

	$\sqrt{16x^2}$
Since $(4x)^2 = 16x^2$	$4x$

TRY IT 5.151 Simplify: $\sqrt{64x^2}$.

TRY IT 5.152 Simplify: $\sqrt{169y^2}$.

EXAMPLE 5.77

Simplify: $-\sqrt{81y^2}$.

Solution

	$-\sqrt{81y^2}$
Since $(9y)^2 = 81y^2$	$-9y$

TRY IT 5.153 Simplify: $-\sqrt{121y^2}$.

TRY IT 5.154 Simplify: $-\sqrt{100p^2}$.

EXAMPLE 5.78

Simplify: $\sqrt{36x^2y^2}$.

Solution

	$\sqrt{36x^2y^2}$
Since $(6xy)^2 = 36x^2y^2$	$6xy$

TRY IT 5.155 Simplify: $\sqrt{100a^2b^2}$.

TRY IT 5.156 Simplify: $\sqrt{225m^2n^2}$.

Use Square Roots in Applications

As you progress through your college courses, you'll encounter several applications of square roots. Once again, if we use our strategy for applications, it will give us a plan for finding the answer!

HOW TO

Use a strategy for applications with square roots.

Step 1. Identify what you are asked to find.
Step 2. Write a phrase that gives the information to find it.
Step 3. Translate the phrase to an expression.
Step 4. Simplify the expression.
Step 5. Write a complete sentence that answers the question.

Square Roots and Area

We have solved applications with area before. If we were given the length of the sides of a square, we could find its area by squaring the length of its sides. Now we can find the length of the sides of a square if we are given the area, by finding the square root of the area.

If the area of the square is A square units, the length of a side is $\sqrt{A}$ units. See Table 5.7.

Area (square units)	Length of side (units)
9	$\sqrt{9}=3$
144	$\sqrt{144}=12$
A	$\sqrt{A}$

Table 5.7

EXAMPLE 5.79

Mike and Lychelle want to make a square patio. They have enough concrete for an area of 200 square feet. To the nearest tenth of a foot, how long can a side of their square patio be?

Solution

We know the area of the square is 200 square feet and want to find the length of the side. If the area of the square is A square units, the length of a side is $\sqrt{A}$ units.

What are you asked to find?	The length of each side of a square patio
Write a phrase.	The length of a side
Translate to an expression.	$\sqrt{A}$
Evaluate $\sqrt{A}$ when $A=200$.	$\sqrt{200}$
Use your calculator.	$14.142135\ldots$

Round to one decimal place.	14.1 feet
Write a sentence.	Each side of the patio should be 14.1 feet.

TRY IT 5.157 Katie wants to plant a square lawn in her front yard. She has enough sod to cover an area of 370 square feet. To the nearest tenth of a foot, how long can a side of her square lawn be?

TRY IT 5.158 Sergio wants to make a square mosaic as an inlay for a table he is building. He has enough tile to cover an area of 2704 square centimeters. How long can a side of his mosaic be?

Square Roots and Gravity

Another application of square roots involves gravity. On Earth, if an object is dropped from a height of h feet, the time in seconds it will take to reach the ground is found by evaluating the expression $\frac{\sqrt{h}}{4}$. For example, if an object is dropped from a height of 64 feet, we can find the time it takes to reach the ground by evaluating $\frac{\sqrt{64}}{4}$.

	$\frac{\sqrt{64}}{4}$
Take the square root of 64.	$\frac{8}{4}$
Simplify the fraction.	2

It would take 2 seconds for an object dropped from a height of 64 feet to reach the ground.

EXAMPLE 5.80

Christy dropped her sunglasses from a bridge 400 feet above a river. How many seconds does it take for the sunglasses to reach the river?

Solution

What are you asked to find?	The number of seconds it takes for the sunglasses to reach the river
Write a phrase.	The time it will take to reach the river
Translate to an expression.	$\frac{\sqrt{h}}{4}$
Evaluate $\frac{\sqrt{h}}{4}$ when $h = 400$.	$\frac{\sqrt{400}}{4}$
Find the square root of 400.	$\frac{20}{4}$
Simplify.	5
Write a sentence.	It will take 5 seconds for the sunglasses to reach the river.

TRY IT 5.159 A helicopter drops a rescue package from a height of 1296 feet. How many seconds does it

take for the package to reach the ground?

TRY IT 5.160 A window washer drops a squeegee from a platform 196 feet above the sidewalk. How many seconds does it take for the squeegee to reach the sidewalk?

Square Roots and Accident Investigations

Police officers investigating car accidents measure the length of the skid marks on the pavement. Then they use square roots to determine the speed, in miles per hour, a car was going before applying the brakes. According to some formulas, if the length of the skid marks is d feet, then the speed of the car can be found by evaluating $\sqrt{24d}$.

EXAMPLE 5.81

After a car accident, the skid marks for one car measured 190 feet. To the nearest tenth, what was the speed of the car (in mph) before the brakes were applied?

Solution

What are you asked to find?	The speed of the car before the brakes were applied
Write a phrase.	The speed of the car
Translate to an expression.	$\sqrt{24d}$
Evaluate $\sqrt{24d}$ when $d = 190$.	$\sqrt{24 \cdot 190}$
Multiply.	$\sqrt{4{,}560}$
Use your calculator.	$67.527772\ldots$
Round to tenths.	67.5
Write a sentence.	The speed of the car was approximately 67.5 miles per hour.

TRY IT 5.161 An accident investigator measured the skid marks of a car and found their length was 76 feet. To the nearest tenth, what was the speed of the car before the brakes were applied?

TRY IT 5.162 The skid marks of a vehicle involved in an accident were 122 feet long. To the nearest tenth, how fast had the vehicle been going before the brakes were applied?

LINKS TO LITERACY

The *Links to Literacy* activity "Sea Squares" will provide you with another view of the topics covered in this section.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Introduction to Square Roots (https://www.openstax.org/l/24introsqroots)

Estimating Square Roots with a Calculator (https://www.openstax.org/l/24estsqrtcalc)

SECTION 5.7 EXERCISES

Practice Makes Perfect

Simplify Expressions with Square Roots

In the following exercises, simplify.

489. $\sqrt{36}$

490. $\sqrt{4}$

491. $\sqrt{64}$

492. $\sqrt{144}$

493. $-\sqrt{4}$

494. $-\sqrt{100}$

495. $-\sqrt{1}$

496. $-\sqrt{121}$

497. $\sqrt{-121}$

498. $\sqrt{-36}$

499. $\sqrt{-9}$

500. $\sqrt{-49}$

501. $\sqrt{9+16}$

502. $\sqrt{25+144}$

503. $\sqrt{9}+\sqrt{16}$

504. $\sqrt{25}+\sqrt{144}$

Estimate Square Roots

In the following exercises, estimate each square root between two consecutive whole numbers.

505. $\sqrt{70}$

506. $\sqrt{55}$

507. $\sqrt{200}$

508. $\sqrt{172}$

Approximate Square Roots with a Calculator

In the following exercises, use a calculator to approximate each square root and round to two decimal places.

509. $\sqrt{19}$

510. $\sqrt{21}$

511. $\sqrt{53}$

512. $\sqrt{47}$

Simplify Variable Expressions with Square Roots

In the following exercises, simplify. (Assume all variables are greater than or equal to zero.)

513. $\sqrt{y^2}$

514. $\sqrt{b^2}$

515. $\sqrt{49x^2}$

516. $\sqrt{100y^2}$

517. $-\sqrt{64a^2}$

518. $-\sqrt{25x^2}$

519. $\sqrt{144x^2y^2}$

520. $\sqrt{196a^2b^2}$

Use Square Roots in Applications

In the following exercises, solve. Round to one decimal place.

521. **Landscaping** Reed wants to have a square garden plot in his backyard. He has enough compost to cover an area of 75 square feet. How long can a side of his garden be?

522. **Landscaping** Vince wants to make a square patio in his yard. He has enough concrete to pave an area of 130 square feet. How long can a side of his patio be?

523. **Gravity** An airplane dropped a flare from a height of $1{,}024$ feet above a lake. How many seconds did it take for the flare to reach the water?

524. **Gravity** A hang glider dropped his cell phone from a height of 350 feet. How many seconds did it take for the cell phone to reach the ground?

525. **Gravity** A construction worker dropped a hammer while building the Grand Canyon skywalk, $4{,}000$ feet above the Colorado River. How many seconds did it take for the hammer to reach the river?

526. **Accident investigation** The skid marks from a car involved in an accident measured 54 feet. What was the speed of the car before the brakes were applied?

527. **Accident investigation** The skid marks from a car involved in an accident measured 216 feet. What was the speed of the car before the brakes were applied?

528. **Accident investigation** An accident investigator measured the skid marks of one of the vehicles involved in an accident. The length of the skid marks was 175 feet. What was the speed of the vehicle before the brakes were applied?

529. **Accident investigation** An accident investigator measured the skid marks of one of the vehicles involved in an accident. The length of the skid marks was 117 feet. What was the speed of the vehicle before the brakes were applied?

Everyday Math

530. **Decorating** Denise wants to install a square accent of designer tiles in her new shower. She can afford to buy 625 square centimeters of the designer tiles. How long can a side of the accent be?

531. **Decorating** Morris wants to have a square mosaic inlaid in his new patio. His budget allows for $2{,}025$ tiles. Each tile is square with an area of one square inch. How long can a side of the mosaic be?

Writing Exercises

532. Why is there no real number equal to $\sqrt{-64}$?

533. What is the difference between 9^2 and $\sqrt{9}$?

Self Check

ⓐ After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.

I can...	Confidently	With some help	No-I don't get it!
simplify expressions with square roots.			
estimate square roots.			
approximate square roots.			
simplify variable expressions with square roots.			
use square roots in applications.			

ⓑ Overall, after looking at the checklist, do you think you are well-prepared for the next Chapter? Why or why not?